Simon's Working Note — Media Archive Audit & Repair Plan

Status date: 2026-11-01
Scope: 2025 Hong Kong / China trip archive (Mar 17 – May 25, 2025)
Status: Audit complete. No repair has been performed yet. This document is the pre-change baseline.

> Everything marked **[verified]** was measured by reading the files or querying the DB on 2026-11-01.
> Everything marked **[assumed]** comes from Simon's recollection and has not been independently proven.

1. Purpose

This note records:

1. What actually lives in each of the four media directories.
2. Simon's copy / collect / delete assumptions, and whether the evidence supports them.
3. The current state of every file as of 2026-11-01 — what is trustworthy and what is damaged.
4. The Phase 0–5 repair plan.

2. The four directories

All four sit on `/Volumes/SSD_HD` (exFAT, 3.6 TB, ~1.9 TB free).

Short names used throughout this note

```
DIR_A = /Volumes/SSD_HD/Keep_Media_files/testing_media_dir_2025-Nov-18
DIR_B = /Volumes/SSD_HD/Keep_Media_files/Videos_Original_Inc_Simon_iPhone
DIR_C = /Volumes/SSD_HD/Keep_Media_files/2025HKTrip_Organized_from_Original
DIR_D = /Volumes/SSD_HD/media
```

Name	Role	Real files	Size (du)	Structure
`DIR_C`	Pristine curated master	3,235	290 GB	23 hand-named folders, flat, **files renamed**
`DIR_B`	Raw archive (`DIR_C` + other people's media + LRF)	4,410	378 GB	31 folders, original names
`DIR_A`	Processed / scanned / de-duplicated working copy	3,403	293 GB	same 31 folders as `DIR_B`

Name	Role	Real files	Size (du)	Structure
`DIR_D`	**Duplicate of `DIR_A`** (no `*.db`) — configured `base_directory` in the DB	3,401	293 GB	same as `DIR_A`

2.1 File inventory [verified]

Extension	`DIR_A`	`DIR_B`	`DIR_C`
HEIC	2,068	2,220	2,055
MP4	945	945	896
JPG	192	123	102
MOV	108	109	91
PNG	88	115	91
LRF (DJI proxies)	0	897	0
INI / DB	2 (`.db`)	1	0
Total real files	**3,403**	**4,410**	**3,235**
AppleDouble `._*` junk	**1,689**	174	0

Logical byte totals: `DIR_A` = 310.9 GB, `DIR_B` = 403.2 GB (incl. 91.9 GB LRF), `DIR_C` = 309.8 GB.

2.2 What each tree really is [verified]

`DIR_C` — pristine master. Files were **renamed** to Chinese descriptive names (`香港_1.HEIC`, `台灣_1.MP4`, `長洲太平清醮_3.HEIC`, `東莞家族晚餐_Menu3.HEIC`). No Zulu `UserComment` was written. 97 files carry Apple's own `UserComment` ("Screenshot"). 1,612 of 3,235 have GPS. Folder set is a hand-curated regrouping, so **path and filename do not align with `DIR_A` or `DIR_B`** — matching must be by content.

> Note: `DIR_C` is *not* perfectly untouched. 21 of its 3,235 files are not byte-identical to anything in `DIR_B`; at least one (`00_2025_iPhone/IMG_8262.HEIC`) has different rotation metadata. Treat `DIR_C` as "best available original", not "certified untouched".

`DIR_B` — raw archive. A superset of `DIR_C`: 3,214 of `DIR_C`'s 3,235 files match a `DIR_B` file exactly on size + mtime (3,212 also match mtime within 2 s). `DIR_B` adds 1,097 items: 897 LRF sidecars, 152 HEIC, 27 PNG, 19 JPG, 1 MOV, `desktop.ini`. Contains the DJI `.LRF` proxies that `DIR_C` and `DIR_A` do not.

`DIR_A` — processed working copy. Produced by `PreImport_exif_file_processor.py` + the scan pipeline, then manually pruned. Same 31 folders as `DIR_B`, extensions upper-cased, 88 `.HEIC` → `.JPG` corrections, `UserComment` stamped on every file, GPS written into 1,604 files, all mtimes rewritten to Nov 2025, 1,689 AppleDouble files created, and the two `*.db` files copied into the root.

`DIR_D` — duplicate of `DIR_A`. Same 3,401 media files as `DIR_A` with identical relative paths and identical sizes; the only difference is `DIR_A` additionally holds `media_organizer.db` and `media_library.db`. Different inodes → a real second copy, not a symlink. **This is ~293 GB of duplication** and is the path recorded in `config.base_directory`.

3. Simon's assumption vs. the evidence

What Simon said [assumed]

> *(directory names normalised to `DIR_x` for readability)*
> `DIR_C` holds my own original media. I duplicated `DIR_C` to `DIR_B`, then collected other people's media into `DIR_B`. I copied `DIR_B` to `DIR_A`. I ran the scan programs against `DIR_A`, then manually removed LRF files and removed duplicate/useless files from `DIR_A`. `DIR_A` should therefore hold the minimal set. Their creation time and Geo may not be correct.

What the evidence says [verified]

Assumption	Verdict	Evidence
`DIR_C` is Simon's own original set	✅ supported	`DIR_C` ⊂ `DIR_B` on content; all `DIR_C` names are Simon-style Chinese descriptive names
`DIR_B` = `DIR_C` + other people's media	✅ supported	3,214/3,235 `DIR_C` files found in `DIR_B`; `DIR_B` adds 165 HEIC / 27 PNG / 19 JPG / 1 MOV beyond `DIR_C`
`DIR_A` = copy of `DIR_B`	✅ supported	3,354 shared folder+stem items; `DIR_A` holds nothing `DIR_B` lacks except the 2 DB files
Scan programs modified `DIR_A`	✅ confirmed	`UserComment` on 3,401/3,401; GPS added to 1,604 files; all mtimes → Nov 2025
LRF files removed from `DIR_A`	✅ confirmed	897 LRF in `DIR_B` (91.9 GB), 0 in `DIR_A`
Duplicates / useless files removed from `DIR_A`	⚠️ **only partly true**	111 media files (328.9 MB) were removed, but **none of them is a content duplicate of anything left in `DIR_A`**. They were unique files.
`DIR_A` is the minimal set	❌ not quite	`DIR_A` keeps 3,401 media items; `DIR_C`'s curated set is 3,235. Also `DIR_A` is duplicated in full at `DIR_D`.
`DIR_A`'s creation times / Geo may be wrong	✅ **confirmed, and quantified below**	1,597 of 3,401 stamped timestamps are provably wrong

3.1 What was actually deleted from `DIR_A`

111 media files, 328.9 MB: **64 HEIC, 27 PNG, 19 JPG, 1 MOV**.

Folder	Count	Nature
`Orders`	52	receipts, screenshots, document photos
`Captures`	17	iOS screenshots
`HK_Bank`	8	screenshots

Folder	Count	Nature
`01_2025Mar香港`	7	travel photos
`Miu_iPhone/202505`	6	photos
`05_2025Apr深圳`	5	mixed
`iPhone`	4	incl. a 135 MB screen recording
others (12 folders)	12	mixed

- 96 of the 111 are still preserved in **`DIR_C`**.
- **15 exist nowhere but `DIR_B`** (41.0 MB) — see §7.

4. Method

1. Full inventory of all four trees (path, size, mtime) → `/tmp/inv.json`.
2. Set comparison by folder + stem (case- and extension-normalised), then by content.
3. **Content verification of every shared item:**
   - Video → MD5 of the `mdat` payload / `ffmpeg -c copy -f md5` (ignores container metadata).
   - Image → MD5 of decoded RGB pixels (ignores EXIF).
   - Result file: `/tmp/content_verify.jsonl` (3,354 records).
4. Metadata survey with `exiftool -j -G1` over all 3,401 files → `/tmp/rich_A.json`.
5. Cross-checks against independent in-file sources (`GPSDateTime`, `Keys:CreationDate`, filename clock).
6. DB introspection with `sqlite3`.

> The scratch artifacts under `/tmp` (`inv.json`, `rich_A.json`, `content_verify.jsonl`) are regenerable and may not survive a reboot; Phase 1 rebuilds an equivalent, permanent ledger inside the project.

Known false positive, corrected: an initial pass flagged 11 Shenzhen MP4s as content-different. Cause was pairing a Live Photo's `.MP4` against its sibling `.HEIC`. After pairing by extension: **48/48 MP4 payloads and 97/97 images identical**.

5. Current status as of 2026-11-01

5.1 Content integrity — CLEAN

All 3,354 media items shared between `DIR_A` and `DIR_B` have identical content. No re-encoding, no resizing, no quality loss anywhere. `DIR_A` is a pure metadata rewrite of `DIR_B`.

5.2 The critical reassurance

No original capture timestamp was ever modified. `QuickTime:CreateDate` is byte-for-byte unchanged in **3,313 / 3,313** shared files. Only a new `UserComment` field was added. Every file still contains its own ground truth.

5.3 Metadata written into `DIR_A`

Change	Scope
`UserComment` = UTC Zulu (`YYYY-MM-DDTHH:MM:SSZ`)	3,401 / 3,401
GPS coordinates written into the file (XMP for video, EXIF for stills)	**+1,604** files (3,223 with GPS vs 1,619 in `DIR_B` ; 0 lost)
Extension case normalised to upper case	all
Real-type correction `.HEIC` → `.JPG` (iOS had mislabelled JPEG data)	88 (`Shenzhen_2025`)
`FileModifyDate` overwritten	3,401 / 3,401 → all `2025-11-17/18`
AppleDouble `._*` junk created	1,689
DB files copied into root	`media_organizer.db` , `media_library.db`

5.4 Timestamp damage — 1,597 provably wrong

Each stamped `UserComment` compared against the file's own authoritative source:

Type	Files	Stamped correct	Provably wrong	Unverifiable	Error pattern
HEIC	2,068	1,530	**538**	0	**+12 h** (all `Miu_iPhone` + 94 in `Shenzhen_2025`)
JPG	192	28	**95**	69	**+12 h** (every one exactly 43,200 s)
MOV	108	88	**20**	0	**+4 h ×19** (all in `Shenzhen_2025`), +51 days ×1 (`63.MOV`), −16 min ×1 (screen recording)
MP4	945	1	**944**	0	**end-of-clip, not start**: median +51 s, p90 330 s, max 14,400 s
PNG	88	0	0	88	no timezone in file — cannot validate
Total	**3,401**	**1,647**	**1,597**	**157**	

5.5 GPS damage — mild

Metric	Count
Files with GPS before (`DIR_B`)	1,619
Files with GPS after (`DIR_A`)	3,223
GPS added	**+1,604**
Original GPS preserved untouched	**1,602 / 1,619**
Original GPS **modified** (needs review)	**17** (e.g. `IMG_8925` : 30°14'54.16"N → 30°14'54.75"N, ≈18 m)
GPS lost	0

5.6 Device and file-type inventory [verified]

Source	Files	Reliable time field	Reliable GPS field
iPhone 15 Plus (Simon)	**1,619** (1,528 HEIC + 88 MOV + 3 JPG)	`DateTimeOriginal` + `OffsetTimeOriginal` ; `Keys:CreationDate`	EXIF GPS; `Keys:GPSCoordinates`
iPhone 13 (wife, `Miu_iPhone`)	**587** (538 HEIC + 49 JPG)	`DateTimeOriginal` + `OffsetTimeOriginal` (all `+08:00`)	EXIF GPS
iPhone 15	**54** (40 JPG + 14 MOV)	as above	as above
iPhone 8 Plus	**2** HEIC	as above	as above
Samsung Galaxy A54 5G	**6** JPG	EXIF	EXIF
DJI Pocket 3 (`PP-101`)	**949** (945 MP4 + 4 JPG)	`QuickTime:CreateDate` = **UTC**	**none**
No model (screenshots / exports)	**184** (90 JPG + 88 PNG + 6 MOV)	none reliable	rarely
Total	**3,401**		

Timezone offsets found: **+08:00 on 2,189 files**, **–04:00 on 2 files only** (`Captures/IMG_0217.HEIC` , `Captures/IMG_0218.HEIC` , 2025-05-25 — after returning to Toronto). 2,191 files carry an EXIF offset in total; the other 1,210 do not (945 MP4 + 108 MOV + 88 PNG + 69 JPG — MP4/MOV carry their time in the container instead).

5.7 Independent verification of the good data

Check	Sample	Result
HEIC `DateTimeOriginal` +offset vs **GPS satellite time** (`GPSDateTime`)	1,515	median **1.0 s**, p99 18 s, 1,505 within 30 s, 4 outliers > 60 s
Own phone vs wife's phone, same day, median latitude	47 shared days	**40 days within 0.05°** (≈5 km); remainder = different parts of the same city
MOV `QuickTime:CreateDate` vs `Keys:CreationDate`	103	**97 exact**; 6 inconsistent
DJI filename clock vs `QuickTime:CreateDate`	8 samples, Mar 16 – May 20	**constant +4:00** → filename = Toronto local, CreateDate = UTC

Conclusion: the iPhone's own EXIF time and GPS are trustworthy — on **both** phones. Only the data our program stamped over them is wrong.

5.8 Known outliers requiring manual review

File	Problem
`IMG_8186.HEIC`	EXIF time is 84 min later than GPS satellite time
`IMG_8450.HEIC`, `IMG_8451.HEIC`, `IMG_9209.HEIC`	62–95 s later than GPS satellite time
`05_2025Apr深圳/63.MOV`	`QuickTime:CreateDate` and `Keys:CreationDate` differ by **51 days**
`IMG_8291.MOV`, `IMG_8460.MOV`, `IMG_8888.MOV`, `IMG_8889.MOV`, `IMG_8943.MOV`	internal timestamps differ by 5–100 min
All 19 `Shenzhen_2025/*.MOV`	stamped 4 hours late (double-applied −04:00 offset); originals are correct in `Keys:CreationDate`
`ScreenRecording_04-10-2025 13-41-35_1.MOV`	stamped −944 s vs authority

5.9 Database contents

`/Users/syuen/Video_editor_Project/Vacaction_Media_Org/media_organizer.db` (the one the web app uses — see `media_library_web/src/main.py`, which points at the repo root):

Table	Rows	Rebuildable?
`journal`	74 (65 with text)	❌ **irreplaceable — Simon's diary**
`geo_data`	4,823	✅ exact 1:1 copy of `geo_chinese.list` (verified: 0 rows differ either way)
`config`	4	✅ trivial
`media_files`	3,401	✅ rebuild from a ledger

`config` values: `OffsetTime = +08:00`, `TimeZone = Asia/Hong_Kong`, `base_directory = /Volumes/SSD_HD/media`, `thumb_directory = /Volumes/SSD_HD/media_thumb`.

`media_files` path column is **relative** (e.g. `01_2025Mar香港/DJI_….MP4`), resolved against `base_directory`.

GPS provenance recorded in `media_files`:

Media	`hasGPS`	`shareGPS`	Count	Meaning
Image	1	0	1,499	native EXIF GPS
Image	0	1	696	inherited from a neighbour
Image	0	0	135	no location
Image	1	1	18	both flags set (anomaly)
Video	0	1	955	inherited
Video	1	0	96	set manually via FixInfo
Video	1	1	2	both flags set (anomaly)

`media_library.db` is a near-empty legacy database (`journal` 2 rows, `media` 2 rows). Safe to ignore.

> ⚠️ Chinese city names in `geo_chinese.list` are machine transliterations with errors, e.g. `Wanning → 减弱` (should be 万宁), `Wuzhen → 武恩` (should be 烏鎮), `Lanyang → 兰南`. Cosmetic, but worth a separate cleanup.

5.10 Root causes (code)

Symptom	Cause	Location
633 stills +12 h wrong	Uses the **file's** `FileInodeChangeDate` offset as if it were the capture timezone. Files copied in Toronto (–04:00) had that offset applied to Asia (`+08:00`) wall-clock times.	`calculate_creation_time()`, `PreImport_exif_file_processor.py:468`
MOV +4 h	Same bug applied to `QuickTime:CreateDate`, which is already UTC	same
MP4 end-of-clip	Falls back to file mtime (written when the clip closes) instead of `QuickTime:CreateDate`	same
PNG unverifiable	No timezone exists in PNG; whatever offset is applied is invented	same
Videos `+duration` in DB	`metadata_extractor.py` prioritises `UserComment`, which is already wrong	`metadata_extractor.py:865`
One global display offset	`+08:00` applied to every row regardless of where it was shot	`media_library_web/src/routes/media.py:179`

The correct authority per file type — always prefer these over mtime:

Type	UTC from
HEIC / JPG	`EXIF:DateTimeOriginal` + `EXIF:OffsetTimeOriginal`
MOV	`Keys:CreationDate` (has offset) → `QuickTime:CreateDate` (UTC)
MP4	`QuickTime:CreateDate` (UTC)
PNG / unknown	**none** — mark as unknown, never guess

6. Answers to the four questions

1. **Can iPhone HEIC time and GPS be trusted?** Yes, on both phones. 100% of 2,068 HEICs carry `DateTimeOriginal` + `OffsetTimeOriginal`; GPS exists on 2,059. Cross-checked against GPS satellite time: median error 1 second. The wife's phone is equally reliable — its original EXIF is intact; only our stamp is wrong.
2. **Can JPG / MOV be trusted?** MOV yes (97/103 internally consistent, 108 with UTC container time, 98 with GPS); 6 files need triage. JPG partly — 123 of 192 carry a timezone, 152 carry GPS; the other ~65 are screenshots or exports with no usable time.

3. **The MP4 modifications need rework?** Yes — 944 of 945 stamps are end-of-clip rather than start. But the fix is safe: the untouched `QuickTime:CreateDate` still holds the true start time in every file.
4. **Is the DB at risk?** No. `geo_data` is a byte-equivalent copy of `geo_chinese.list`; only the 74-row `journal` is irreplaceable and needs an export before any rebuild.

7. Files that exist only in `DIR_B` (at risk)

15 files, 41.0 MB. Present in `DIR_B`; absent from `DIR_A` and from `DIR_C`.

```
01_2025Mar香港/IMG_8183.HEIC        3,005,197
01_2025Mar香港/IMG_8189.HEIC        3,196,541
01_2025Mar香港/IMG_8190.HEIC        2,325,927
01_2025Mar香港/IMG_8191.HEIC        2,354,540
01_2025Mar香港/IMG_8235.HEIC        2,601,412
01_2025Mar香港/IMG_8320.HEIC        2,613,098
05_2025Apr深圳/IMG_8294.PNG           629,590
05_2025Apr深圳/IMG_8295.HEIC        2,504,065
10_2025Apr東莞/IMG_8566.HEIC        2,999,075
40_2025Apr_/IMG_9062.PNG            4,225,508
Miu_iPhone/202505/IBHS3955.JPG      2,165,124
Miu_iPhone/202505/IMG_1836.HEIC     2,334,021
Miu_iPhone/202505/IMG_1903.HEIC     2,909,484
Miu_iPhone/202505/IMG_1907.HEIC     1,783,300
iPhone/IMG_9748.HEIC                5,319,456
```

`DIR_B` must not be deleted until these 15 are restored to `DIR_C` or `DIR_A`.

8. Repair plan

Guiding principles

1. **`DIR_B` and `DIR_C` are read-only.** No process may write to them.
2. **Never re-run PreImport against `DIR_A`.** `DIR_A`'s mtimes are destroyed, so any recomputation is guaranteed wrong.
3. **The files already hold the truth.** Repair means *reading* the authoritative field and *replacing derived data*, not inventing values.
4. **Derived data belongs in the DB, not in the media files.** Inheritance and manual corrections should live in SQLite; files should stay as the camera wrote them.
5. **Every derived value carries provenance and confidence.**

Phase 0 — Freeze and back up

- ☐ Make `DIR_B` and `DIR_C` read-only; never write to them.
- ☐ `sqlite3 media_organizer.db ".backup media_organizer.db.2026-11-01.bak"`

- [ ] Export the journal:

```
sqlite3 media_organizer.db ".mode insert journal" "SELECT * FROM journal;" > journal_export_2026-11-01.sql
```

- [ ] Snapshot `geo_chinese.list` checksum (it is the true source of `geo_data`).
- [ ] Stop running `PreImport_exif_file_processor.py` and `Main_scan_media.py` against `DIR_A` or `DIR_D`.
- [ ] Decide the fate of `DIR_D` (293 GB duplicate of `DIR_A`).

Exit criteria: backups exist and verify; `DIR_B` and `DIR_C` untouched; no pipeline runs scheduled.

Phase 1 — Build the ledger (read-only, changes nothing)

Create `build_media_ledger.py` producing `media_ledger` (SQLite table + CSV export), one row per media file in `DIR_A`:

Column	Rule
`relpath`, `ext`, `size`, `device`	from the filesystem + EXIF `Make` / `Model`
`capture_utc`	HEIC/JPG: `DateTimeOriginal` + `OffsetTimeOriginal` · MOV: `Keys:CreationDate` → `QuickTime:CreateDate` · MP4: `QuickTime:CreateDate` · PNG: `NULL`
`time_source`	`exif_offset` / `keys` / `quicktime` / `filename` / `unknown`
`time_confidence`	`high` / `medium` / `unknown`
`lat`, `lon`	EXIF GPS → `Keys:GPSCoordinates` → `NULL`
`geo_source`	`exif` / `keys` / `inherited:<file>` / `manual` / `none`
`old_usercomment`	the current (often wrong) stamp, kept for audit
`stamp_error_s`	`old_usercomment − capture_utc` in seconds

Expected outcome: **3,244 files with authoritative UTC** (2,068 HEIC + 123 JPG + 108 MOV + 945 MP4) and **157 unknown** (88 PNG + 69 JPG). For GPS: **1,619 native** coordinates survive from the originals; the remaining ~1,650 currently hold inherited or hand-entered values that must be re-derived (see Phase 3) rather than trusted. Note the DB and the files are slightly out of sync — files show 3,223 with GPS, the DB's own flags imply 1,615 native + 1,651 inherited + 135 none — which is one more reason to rebuild from a single ledger.

Exit criteria: Simon reviews the CSV and agrees with the counts. Nothing on disk has changed.

Phase 2 — Restore files that should never have changed

2a. 6 internally inconsistent MOVs — copy back from `DIR_B`:

```
05_2025Apr深圳/63.MOV
05_2025Apr深圳/IMG_8291.MOV
05_2025Apr深圳/IMG_8460.MOV
25_2025Apr上海/IMG_8888.MOV
25_2025Apr上海/IMG_8889.MOV
40_2025Apr_/IMG_8943.MOV
```

2b. 17 photos whose original EXIF GPS was overwritten — copy back from `DIR_B` (their original coordinates are intact there):

File	Original (`DIR_B`)	Currently in `DIR_A`
`IMG_8240.HEIC`	22°17'23.96"N	22°17'21.28"N
`IMG_8241.HEIC`	22°17'23.96"N	22°17'21.28"N
`IMG_8297.HEIC`	22°17'04.10"N	22°16'39.79"N (≈750 m)
`IMG_8446.HEIC`	22°32'37.79"N	22°32'38.73"N
`IMG_8447.HEIC`	22°32'35.89"N	22°32'38.73"N
`IMG_8793.HEIC`	31°14'16.03"N	31°06'26.94"N (**≈14 km**)
`IMG_8925.HEIC`	30°14'54.16"N	30°14'54.75"N
`IMG_8926.HEIC`	30°14'53.50"N	30°14'54.75"N
`IMG_8952.HEIC`	31°54'55.59"N	31°54'56.10"N
`IMG_8953.HEIC`	31°54'55.46"N	31°54'56.10"N
`IMG_8954.HEIC`	31°54'55.46"N	31°54'56.10"N
`IMG_8955.HEIC`	31°54'55.46"N	31°54'56.10"N
`IMG_9135.HEIC`	30°41'12.19"N	30°41'11.61"N
`IMG_9136.HEIC`	30°41'11.97"N	30°41'11.61"N
`IMG_9149.HEIC`	30°41'17.03"N	30°41'11.61"N
`IMG_9233.HEIC`	31°14'16.49"N	31°14'16.59"N
`IMG_9234.HEIC`	31°14'16.47"N	31°14'16.59"N

Most are tens of metres — consistent with "snap to saved location" bulk edits. `IMG_8793` (14 km) and `IMG_8297` (750 m) look like genuine mistakes and must be reverted. **Confirm with Simon before reverting the small ones**, which may be deliberate corrections.

2c. 15 files that currently exist only in `DIR_B` (§7) — restore into `DIR_C` first, then into `DIR_A` .

Exit criteria: `DIR_A` 's GPS values match `DIR_B` 's for every file that had GPS in `DIR_B` (or a documented, deliberate override is recorded); the 15 orphans are preserved in at least two locations.

Phase 3 — Fix the derived metadata

Decision required from Simon — choose one:

- **Option 1 (recommended): files stay as the camera wrote them.** Strip the program-written `UserComment` and the program-added GPS. All derived values live in the DB only. Cleanest for archival integrity.
- **Option 2 (PhotoPrism-friendly): rewrite `UserComment` with the *correct* UTC** from the ledger, and keep GPS in files only where the source is `exif` or `manual` (not `inherited`).

For the 945 DJI clips, re-run GPS inheritance properly:

- ☐ Match on **clip start UTC**, not end.
- ☐ Hard window 20 min; widen to 45 min only if no candidate exists.
- ☐ Require a geo-tagged photo **before and after** the clip that agree within 1 km; otherwise mark `confidence = low`.
- ☐ Reject any match implying travel > 300 km/h versus the previous confirmed anchor.
- ☐ Fall back to the day's location cluster (majority vote) rather than a single nearest photo.
- ☐ Manually review the 96 clips with `hasGPS=1` (manual entries) and the 142 clips whose coordinates match no photo.

Exit criteria: 0 timestamp stamps provably disagree with `capture_utc`; every inherited coordinate carries a confidence label.

Phase 4 — Rebuild the DB

- ☐ Preserve `journal` (74 rows) and `config` (4 rows).
- ☐ Verify `geo_data` still equals `geo_chinese.list`; then reload it to be safe.
- ☐ Recreate `media_files` from `media_ledger`, adding: `capture_utc`, `time_source`, `time_confidence`, `geo_source`, `geo_confidence`.
- ☐ Keep `creation_time` as the UTC string the web app expects, so **no front-end change is needed**.
- ☐ Fix the display logic: convert each row using its **own** `timezone` (from `geo_data`), falling back to `config.OffsetTime` only when the row has none.

Exit criteria: the web pages render identically to today for Asia-shot media, and Toronto-shot media now display in Toronto time instead of being shifted to +08:00.

Phase 5 — Prove the repair

Re-run these checks and require:

Check	Acceptance
HEIC `DateTimeOriginal` +offset vs `GPSDateTime`	median < 2 s; report any file > 60 s
MOV `QuickTime:CreateDate` vs `Keys:CreationDate`	0 mismatches (or all documented)
MP4 filename clock + 4:00 vs `QuickTime:CreateDate`	0 mismatches
`UserComment` (if retained) vs `capture_utc`	0 mismatches
Content hash `DIR_A` vs `DIR_B`	still 3,354/3,354 identical

Check	Acceptance
Row count	`media_files` = `media_ledger` = actual file count
Journal	74 rows, byte-identical text to the Phase 0 export
Web UI spot-check	10 known dates display the expected local times and locations

9. Open decisions

#	Decision	Options
1	What to do with `DIR_D` (293 GB duplicate of `DIR_A`)	keep as scan target · delete · make it the only working copy
2	Phase 3 option	leave files untouched (Option 1) vs write corrected UTC+GPS for PhotoPrism (Option 2)
3	Keep or drop the 88 PNG screenshots and ~65 JPGs with no usable date	drop · keep in a separate `Screenshots/` folder
4	Rename strategy for `DIR_A`	keep original camera names · adopt `DIR_C`'s Chinese descriptive names
5	Whether to clean the machine-translated Chinese city names in `geo_chinese.list`	now · later · never

10. Reference — commands used

```
# inventory (path, size, mtime)
find "$DIR" -type f ! -name ".*" -exec stat -f '%N|%z|%m' {} +

# authoritative time, per file type
exiftool -j -G1 -FileName -Make -Model \
  -EXIF:DateTimeOriginal -EXIF:OffsetTimeOriginal \
  -QuickTime:CreateDate -Keys:CreationDate \
  -EXIF:GPSLatitude -EXIF:GPSLongitude -Keys:GPSCoordinates \
  "$FILE"

# video content check (ignores container metadata)
ffmpeg -v error -i "$FILE" -map 0 -c copy -f md5 -

# DJI reality check: filename clock is Toronto local, CreateDate is UTC
exiftool -s3 -CreateDate "$DJI_MP4"      # 2025:03:20 08:39:32
# filename DJI_20250320043931 → 04:39:31 local (UTC-4) → 08:39:31Z  (+4:00)

# DB integrity
sqlite3 media_organizer.db "SELECT media_type,hasGPS,shareGPS,COUNT(*) FROM media_files GROUP BY 1,2,3;"
sqlite3 media_organizer.db ".backup media_organizer.db.bak"
```

End of note. Next action: Simon reviews Phase 1's expected counts and answers the Open Decisions, then implementation begins.